

Likelihood Assignment 2

Question 1

By conditional independence of the T_i 's given λ ,

$$\pi(t \mid \lambda) = \prod_{i=1}^n \pi(t_i \mid \lambda_i).$$

Also, since the λ_i 's are iid given $\theta = (\alpha, \beta)$,

$$\pi(\lambda \mid \theta) = \prod_{i=1}^n \pi(\lambda_i \mid \theta).$$

Therefore,

$$\pi(t \mid \lambda) \pi(\lambda \mid \theta) = \prod_{i=1}^n \pi(t_i \mid \lambda_i) \pi(\lambda_i \mid \theta).$$

Each term in the product depends only on one λ_i , so the integral factorises:

$$\begin{aligned} L(\theta) &= \int \prod_{i=1}^n \pi(t_i \mid \lambda_i) \pi(\lambda_i \mid \theta) d\lambda_1 \cdots d\lambda_n \\ &= \prod_{i=1}^n \int_0^\infty \pi(t_i \mid \lambda_i) \pi(\lambda_i \mid \theta) d\lambda_i. \end{aligned}$$

Question 3

For repeated visits to café (i), the model is:

$$T_{ij} \mid \lambda_i \sim \text{Exponential}(\lambda_i), \quad j = 1, \dots, v_i, \quad i = 1, \dots, n.$$

The café-specific expected waiting times are modelled as

$$\lambda_i \stackrel{iid}{\sim} \text{Gamma}(\alpha, \beta), \quad i = 1, \dots, n.$$

Conditional on λ_i , the waiting times from repeated visits to the same café are assumed independent.

Question 4

Let $t_i = (t_{i1}, \dots, t_{iv_i})$ denote the waiting times observed at café (i).

The likelihood is obtained by marginalising over the unobserved λ_i 's:

$$L(\theta) = \pi(t \mid \theta) = \int \pi(t \mid \lambda) \pi(\lambda \mid \theta) d\lambda.$$

Given λ , observations from different cafés are independent, so

$$\pi(t \mid \lambda) = \prod_{i=1}^n \pi(t_i \mid \lambda_i).$$

Within café (i), repeated visits are independent conditional on λ_i , hence

$$\pi(t_i \mid \lambda_i) = \prod_{j=1}^{v_i} \pi(t_{ij} \mid \lambda_i).$$

As before, the λ_i 's are iid, therefore:

$$\pi(\lambda \mid \theta) = \prod_{i=1}^n \pi(\lambda_i \mid \theta).$$

Therefore,

$$L(\theta) = \int \prod_{i=1}^n \left[\left\{ \prod_{j=1}^{v_i} \pi(t_{ij} \mid \lambda_i) \right\} \pi(\lambda_i \mid \theta) \right] d\lambda_1 \cdots d\lambda_n.$$

Since each bracketed term depends only on one λ_i , the integral separates:

$$L(\theta) = \prod_{i=1}^n \int_0^\infty \left[\prod_{j=1}^{v_i} \pi(t_{ij} \mid \lambda_i) \right] \pi(\lambda_i \mid \theta) d\lambda_i.$$

Question 5

```
mu <- mean(coffeetime$time)
sigma <- sd(coffeetime$time)

alpha_prior <- mu^{2}/ sigma^{2}
beta_pror <- mu/ sigma^2
```

It is expected that the average café waiting time will be around 4.4 minutes, with substantial variation between cafés, perhaps corresponding to a standard deviation of roughly 5 minutes. Matching these moments to a Gamma distribution gives approximate hyperparameter values of $\alpha \approx 0.78$ and $\beta \approx 0.18$.

Therefore the following exponential priors are placed on the Gamma hyperparameters:

$$\alpha \sim \text{Exponential}\left(\frac{1}{0.78}\right), \quad \beta \sim \text{Exponential}\left(\frac{1}{0.18}\right).$$

These priors are relatively weakly informative, while still centering the model around plausible waiting times and between-café variability.

Question 6

```
random_effects_stan <- "
data {
  int<lower=0> N;
  vector<lower=0>[N] t;
  int<lower=1> n_cafes;
  array[N] int<lower=1, upper=n_cafes> cafe;
```

```

}

parameters {
  real<lower=0> alpha;
  real<lower=0> beta;
  vector<lower=0>[n_cafes] lambda;
}

model {
  alpha ~ exponential(1 / 0.7824712);
  beta ~ exponential(1 / 0.1764025);

  lambda ~ gamma(alpha, beta);

  for (k in 1:N) {
    t[k] ~ exponential(1 / lambda[cafe[k]]);
  }
}
"

writeLines(random_effects_stan, "random_effects_model.stan")

model <- cmdstan_model("random_effects_model.stan")

```

```

stan_data <- list(
  N = length(coffeetime$cafe),
  n_cafes = 15,
  t = coffeetime$time,
  cafe = coffeetime$cafe
)

```

```

fit <- model$sample(
  data = stan_data,
  chains = 4,
  iter_warmup = 1000,
  iter_sampling = 1000,
  seed = 123
)

```

Running MCMC with 4 sequential chains...

```

Chain 1 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 1 Iteration:   100 / 2000 [  5%] (Warmup)

Chain 1 Iteration:   200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration:   300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration:   400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration:   500 / 2000 [ 25%] (Warmup)
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Chain 1 Iteration:  1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration:  1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration:  1100 / 2000 [ 55%] (Sampling)
Chain 1 Iteration:  1200 / 2000 [ 60%] (Sampling)
Chain 1 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 1 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 1 Iteration:  1500 / 2000 [ 75%] (Sampling)
Chain 1 Iteration:  1600 / 2000 [ 80%] (Sampling)
Chain 1 Iteration:  1700 / 2000 [ 85%] (Sampling)
Chain 1 Iteration:  1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration:  1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration:  2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.6 seconds.
Chain 2 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 2 Iteration:   100 / 2000 [  5%] (Warmup)
Chain 2 Iteration:   200 / 2000 [ 10%] (Warmup)

Chain 2 Iteration:   300 / 2000 [ 15%] (Warmup)
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Chain 2 Iteration:  1300 / 2000 [ 65%] (Sampling)
Chain 2 Iteration:  1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration:  1500 / 2000 [ 75%] (Sampling)

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Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)
Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.5 seconds.
Chain 3 Iteration:    1 / 2000 [  0%] (Warmup)
Chain 3 Iteration:  100 / 2000 [  5%] (Warmup)
Chain 3 Iteration:  200 / 2000 [ 10%] (Warmup)

Chain 3 Iteration:  300 / 2000 [ 15%] (Warmup)
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Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.5 seconds.
Chain 4 Iteration:    1 / 2000 [  0%] (Warmup)
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Chain 4 Iteration: 1000 / 2000 [ 50%] (Warmup)

```

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Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.6 seconds.
```

All 4 chains finished successfully.
Mean chain execution time: 0.6 seconds.
Total execution time: 3.1 seconds.

```
sum <- fit$summary()

sum[] <- lapply(sum, function(x) {
  if(is.numeric(x)) round(x, 3) else x
})

kableExtra::kable(sum)
```

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
lp____	-	-	2.934	2.874	-	-	1.002	1306.711	2438.879
	256.580	256.303			261.920	252.352			
alpha	2.043	1.958	0.697	0.673	1.058	3.311	1.001	3429.700	2796.991
beta	0.398	0.374	0.159	0.149	0.180	0.686	1.000	3195.423	2888.306
lambda[1]	5.268	5.026	1.673	1.479	3.099	8.416	1.002	8281.008	2922.799
lambda[2]	6.371	5.639	3.297	2.704	2.617	12.586	1.000	8323.502	2949.979
lambda[3]	3.985	3.720	1.423	1.260	2.204	6.618	1.001	7126.989	3154.319
lambda[4]	7.227	6.657	2.952	2.435	3.690	12.655	1.000	6064.349	3006.705
lambda[5]	5.319	4.950	1.956	1.658	2.923	8.915	1.001	7847.020	2940.066
lambda[6]	8.023	7.564	2.587	2.272	4.685	12.941	1.002	6706.343	3146.934
lambda[7]	2.717	2.441	1.334	1.101	1.232	5.105	1.002	7507.994	3033.447
lambda[8]	1.478	1.255	0.893	0.625	0.565	3.193	1.001	4963.669	2328.570
lambda[9]	4.985	4.715	1.594	1.423	2.948	7.890	1.003	7263.007	2834.421
lambda[10]	8.796	8.200	3.324	2.806	4.761	15.150	1.001	6772.198	2847.472
lambda[11]	3.751	3.468	1.468	1.180	1.971	6.563	1.001	7272.290	2655.411

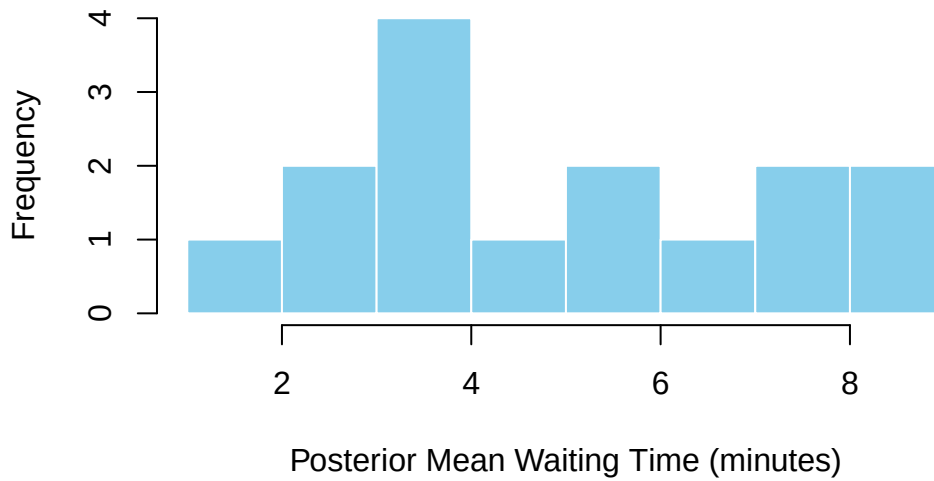
variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
lambda[12]	3.441	3.253	1.127	1.048	1.993	5.561	1.001	7120.898	3143.617
lambda[13]	3.770	3.528	1.299	1.118	2.150	6.150	0.999	8416.608	2793.534
lambda[14]	7.008	6.549	2.448	2.288	3.793	11.616	1.000	8035.386	3033.367
lambda[15]	2.975	2.654	1.431	1.138	1.356	5.654	1.001	7314.221	2810.097

The MCMC diagnostics suggest that the random effects model has fitted successfully. The $\hat{R}$ values are close to 1, which suggests that the chains have mixed well and are sampling from the same posterior distribution. The effective sample sizes are also reasonably large, meaning that the posterior summaries should be reliable. There is therefore no obvious evidence of non-convergence or poor mixing.

The posterior distribution of the café-specific expected waiting times shows that there is variation between cafés. This supports the use of a random effects model, since the model allows each café to have its own expected waiting time while still assuming that the cafés come from a common population distribution.

```
hist(
  sum$mean[4:18],
  main = "Posterior Mean Waiting Times by Café",
  xlab = "Posterior Mean Waiting Time (minutes)",
  ylab = "Frequency",
  border = "white",
  col = "skyblue",
  breaks = 8
)
```


Posterior Mean Waiting Times by Café

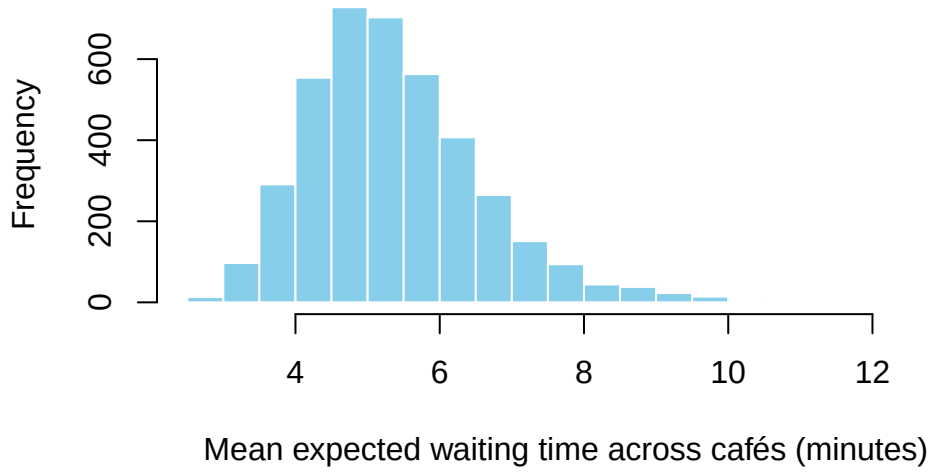


```
samples <- fit$draws(c("alpha", "beta"), format = "matrix")

mean_waiting_time <- samples[, "alpha"] / samples[, "beta"]

hist(
  mean_waiting_time,
  main = "Posterior Draws of the Mean Expected Waiting Time Across Cafés",
  xlab = "Mean expected waiting time across cafés (minutes)",
  ylab = "Frequency",
  breaks = 30,
  col = "skyblue",
  border = "white"
)
```

Posterior Draws of the Mean Expected Waiting Time Across Cafés



Question 7

For the fixed effects model, we still allow the expected waiting time to differ by café, but we do not assume that the café-specific parameters are drawn from a shared Gamma distribution. Thus, there is no pooling of information between cafés.

Let T_{ij} be the waiting time for visit j to café i . I specify $T_{ij} \mid \lambda_i \sim \text{Exponential}(\lambda_i)$, $j = 1, \dots, v_i$, $i = 1, \dots, n$

so that $(T_{ij} \mid \lambda_i) = \lambda_i$.

For the café-specific expected waiting times, independent priors are used:

$$\lambda_i \stackrel{\text{ind}}{\sim} \text{Exponential}(4.4) \quad i = 1, \dots, n$$

I chose 4.4 because in the random effects model, I expected the average café waiting time to be around 4.4 minutes. However, unlike the random effects model, this fixed effects model does not estimate shared parameters such as α and β . Therefore the cafés are not linked through a common estimated population distribution and the posterior estimate of each λ_i is informed only by the data from only café i and its own prior.

Q8

```
fixed_effects_stan <- "  
data {  
  int<lower=0> N;  
  vector<lower=0>[N] t;  
  int<lower=1> n_cafes;  
  array[N] int<lower=1, upper=n_cafes> cafe;  
}  
  
parameters {  
  vector<lower=0>[n_cafes] lambda;  
}  
  
model {  
  lambda ~ exponential(1 / 4.4);  
  
  for (k in 1:N) {  
    t[k] ~ exponential(1 / lambda[cafe[k]]);  
  }  
}  
"  
  
writeLines(fixed_effects_stan, "fixed_effects_model.stan")  
  
fixed_mod <- cmdstan_model("fixed_effects_model.stan")
```

```
library(tidyverse)  
  
# Fit model  
fixed_fit <- fixed_mod$sample(  
  data = stan_data,  
  chains = 4,  
  iter_warmup = 1000,  
  iter_sampling = 1000,  
  seed = 123  
)
```

Running MCMC with 4 sequential chains...

Chain 1 Iteration: 1 / 2000 [0%] (Warmup)

```

Chain 1 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 1 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 1 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 1 Iteration: 400 / 2000 [ 20%] (Warmup)
Chain 1 Iteration: 500 / 2000 [ 25%] (Warmup)

Chain 1 Iteration: 600 / 2000 [ 30%] (Warmup)
Chain 1 Iteration: 700 / 2000 [ 35%] (Warmup)
Chain 1 Iteration: 800 / 2000 [ 40%] (Warmup)
Chain 1 Iteration: 900 / 2000 [ 45%] (Warmup)
Chain 1 Iteration: 1000 / 2000 [ 50%] (Warmup)
Chain 1 Iteration: 1001 / 2000 [ 50%] (Sampling)
Chain 1 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 1 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 1 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 1 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 1 finished in 0.3 seconds.
Chain 2 Iteration: 1 / 2000 [ 0%] (Warmup)
Chain 2 Iteration: 100 / 2000 [ 5%] (Warmup)
Chain 2 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 2 Iteration: 300 / 2000 [ 15%] (Warmup)
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Chain 2 Iteration: 1400 / 2000 [ 70%] (Sampling)
Chain 2 Iteration: 1500 / 2000 [ 75%] (Sampling)
Chain 2 Iteration: 1600 / 2000 [ 80%] (Sampling)

```

```

Chain 2 Iteration: 1700 / 2000 [ 85%] (Sampling)
Chain 2 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 2 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 2 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 2 finished in 0.3 seconds.
Chain 3 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 3 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 3 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 3 Iteration: 300 / 2000 [ 15%] (Warmup)
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Chain 3 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 3 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 3 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 3 finished in 0.3 seconds.
Chain 4 Iteration:   1 / 2000 [  0%] (Warmup)
Chain 4 Iteration: 100 / 2000 [  5%] (Warmup)
Chain 4 Iteration: 200 / 2000 [ 10%] (Warmup)
Chain 4 Iteration: 300 / 2000 [ 15%] (Warmup)
Chain 4 Iteration: 400 / 2000 [ 20%] (Warmup)

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Chain 4 Iteration: 1001 / 2000 [ 50%] (Sampling)

```

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Chain 4 Iteration: 1100 / 2000 [ 55%] (Sampling)
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Chain 4 Iteration: 1800 / 2000 [ 90%] (Sampling)
Chain 4 Iteration: 1900 / 2000 [ 95%] (Sampling)
Chain 4 Iteration: 2000 / 2000 [100%] (Sampling)
Chain 4 finished in 0.3 seconds.
```

```
All 4 chains finished successfully.
Mean chain execution time: 0.3 seconds.
Total execution time: 1.8 seconds.
```

```
# Check summary
fixed_fit$summary()
```

```
# A tibble: 16 x 10
```

	variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
1	lp__	-232.	-232.	2.67	2.66	-237.	-228.	1.00	1505.	2192.
2	lambda[~	5.17	4.89	1.61	1.47	3.09	8.19	1.00	9501.	2859.
3	lambda[~	6.40	5.56	3.65	2.86	2.44	13.5	1.000	8338.	2911.
4	lambda[~	3.86	3.58	1.44	1.26	2.09	6.68	1.00	8899.	3175.
5	lambda[~	7.44	6.78	3.14	2.58	3.64	13.6	1.00	9331.	2869.
6	lambda[~	5.20	4.86	1.89	1.73	2.80	8.77	1.000	9391.	3085.
7	lambda[~	8.33	7.83	2.81	2.45	4.69	13.7	1.00	10307.	3042.
8	lambda[~	2.43	2.11	1.29	0.930	1.07	4.82	1.00	8525.	2863.
9	lambda[~	1.21	1.02	0.720	0.501	0.498	2.51	1.00	8059.	3074.
10	lambda[~	4.88	4.57	1.64	1.38	2.86	7.82	1.00	8281.	2907.
11	lambda[~	9.37	8.65	3.82	3.34	4.74	16.5	1.00	8756.	2847.
12	lambda[~	3.53	3.23	1.39	1.17	1.86	6.15	1.00	8526.	2748.
13	lambda[~	3.29	3.09	1.11	0.941	1.92	5.37	1.00	8765.	3143.
14	lambda[~	3.64	3.37	1.33	1.10	2.04	6.08	1.00	8582.	2577.
15	lambda[~	7.18	6.67	2.66	2.23	3.94	12.4	1.00	9929.	2715.
16	lambda[~	2.72	2.40	1.41	1.03	1.21	5.31	1.00	8526.	2599.

```
Checking MCMC for Q8
```

variable	mean	median	sd	mad	q5	q95	rhat	ess_bulk	ess_tail
lp____	-	-	2.667	2.662	-	-	1.002	1505.291	2191.916
	232.220	231.966			236.893	228.307			
lambda[1]	5.171	4.895	1.608	1.465	3.093	8.192	1.000	9501.021	2858.773
lambda[2]	6.403	5.563	3.649	2.855	2.438	13.493	1.000	8337.908	2911.469
lambda[3]	3.860	3.576	1.443	1.256	2.086	6.680	1.001	8899.171	3174.960
lambda[4]	7.438	6.778	3.141	2.580	3.638	13.606	1.001	9330.747	2869.368
lambda[5]	5.204	4.858	1.890	1.729	2.799	8.769	1.000	9391.456	3084.689
lambda[6]	8.329	7.826	2.811	2.451	4.694	13.660	1.001	10307.375	3042.426
lambda[7]	2.431	2.112	1.288	0.930	1.073	4.824	1.001	8524.826	2863.045
lambda[8]	1.212	1.020	0.720	0.501	0.498	2.514	1.000	8059.389	3073.521
lambda[9]	4.880	4.575	1.643	1.380	2.856	7.821	1.001	8280.555	2906.545
lambda[10]	9.373	8.645	3.821	3.340	4.743	16.475	1.003	8755.949	2846.625
lambda[11]	3.528	3.231	1.388	1.168	1.861	6.149	1.001	8526.147	2748.313
lambda[12]	3.295	3.089	1.114	0.941	1.923	5.367	1.000	8764.533	3142.796
lambda[13]	3.640	3.375	1.327	1.102	2.038	6.081	1.001	8582.357	2576.795
lambda[14]	7.183	6.674	2.658	2.229	3.936	12.374	1.000	9929.176	2714.763
lambda[15]	2.719	2.397	1.407	1.030	1.207	5.306	1.003	8526.259	2599.473

The MCMC diagnostics suggest that the model converged well. All ($\hat{R}$) values are extremely close to 1, indicating good mixing and convergence across chains. In addition, both the bulk and tail effective sample sizes are large for all parameters, suggesting low autocorrelation and reliable estimation of posterior summaries and credible intervals. Therefore, there is no evidence of concerning convergence or sampling issues in the fitted model.

Q9

```
#| echo: true
#| message: false
#| warning: false

# Random effects sum
rand_sum <- fit$summary()

rand_cafe <- rand_sum |>
  filter(str_detect(variable, "^lambda\\[\")) |>
  mutate(
    cafe = as.numeric(str_extract(variable, "\\d+")),
    model = "Random effects"
```

```

) |>
select(cafe,model, median,q5,q95)

# Fixed effects sum
fixed_sum <- fixed_fit$summary()

fixed_cafe <- fixed_sum |>
  filter(str_detect(variable, "^lambda\\[") |>
  mutate(
    cafe = as.numeric(str_extract(variable, "\\d+")),
    model = "Fixed effects"
  ) |>
  select(cafe,model,median,q5,q95)

# Posterior median of overall mean expect waiting time
rand_draws <- fit$draws(c("alpha", "beta"), format = "df")

mu_draws <- rand_draws$alpha / rand_draws$beta
mu_median <- median(mu_draws)

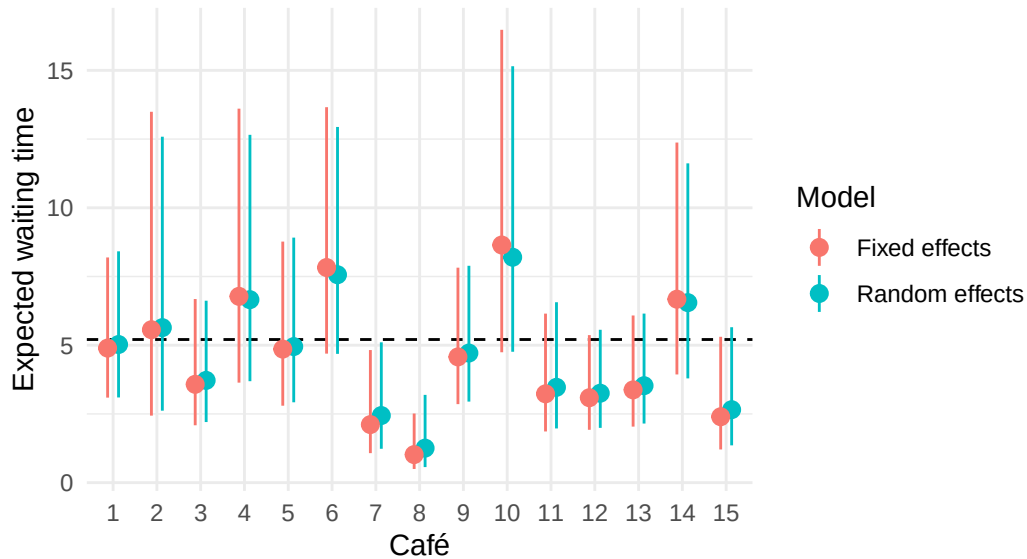
plot_data <- bind_rows(rand_cafe, fixed_cafe)

ggplot(plot_data, aes(x = factor(cafe), y = median, colour = model)) +
  geom_hline(
    yintercept = mu_median,
    linetype = "dashed",
    colour = "black"
  ) +
  geom_pointrange(
    aes(ymin = q5, ymax = q95),
    position = position_dodge(width = 0.5)
  ) +
  labs(
    x = "Café",
    y = "Expected waiting time",
    colour = "Model",
    title = "Comparison of fixed effects and random effects estimates",
    subtitle = "Posterior medians and 90% credible intervals"
  ) +
  theme_minimal()

```


Comparison of fixed effects and random effects estimates

Posterior medians and 90% credible intervals



The plot shows the general difference between the fixed effects and random effects models. The fixed effects model estimates each café separately, so its café-specific estimates are based mainly on the data from that café alone. As a result, the estimates can vary more strongly between cafés, especially for cafés with less data.

The random effects model partially pools information across cafés. This means that café-specific estimates are pulled towards the overall mean expected waiting time. Therefore, cafés with unusually high or low fixed effects estimates tend to have less extreme estimates under the random effects model. The random effects credible intervals may also be narrower or more stable because the model borrows strength across cafés.

The horizontal dashed line represents the posterior median of the overall mean expected waiting time from the random effects model. The random effects café estimates are generally shrunk towards this line, while the fixed effects estimates are allowed to move more freely away from it. This pattern is the main evidence of partial pooling in the random effects model.

Q10

In standard asymptotic theory, standard errors of parameter estimates typically shrink at a rate of $\frac{1}{\sqrt{n}}$ as the sample size increases. In this context, this means that the standard errors will initially decrease as we gain more information.

However, the mean waiting time parameter is designed to describe the whole population of cafés, and continuously sampling from only one café does not provide sufficient information

about the population as a whole. While repeated visits give increasingly precise information about Café Statistique itself, they provide little additional information about between-café variability. Therefore, the standard error for the population mean waiting time will decrease initially, but will not shrink to zero as n becomes very large, and will instead plateau.